| CHEM2201 | PHYSICAL CHEMISTRY-I | L | T | P | S | J | C |
|---|---|---|---|---|---|---|---|
| | | 3 | 0 | 2 | 0 | 0 | 4 |
| Pre-requisite | 10+2 level | | | | | | |
| Co-requisite | Basic Mathematic | | | | | | |
| Preferable Exposure | None | | | | | | |

**Course Description:**

This course introduces students to the core area of physical chemistry, based around the themes of systems, states and processes. Topics covered are chemical thermodynamics, phase changes, chemical kinetics and electrochemistry. Throughout the course, the relationship between physical phenomena and the molecular structure and reactions will be highlighted. The laboratory component provides training in a range of theoretical and applied physical chemistry techniques which are relevant to both industrial and research settings.

**Course Educational Objectives:**

- To demonstrate how thermodynamics dictates the feasibility of physical transformations and chemical reactions
- To provide insights into the chemical kinetics and catalysis, the concept of chemical equilibrium and factors influencing equilibrium
- To introduce the idea of phase transitions of pure substances and phase diagrams
- To generate an intuitive understanding among the students for the concept of Electrochemistry
- Students learn EMF measurements-potentiometric titrations and measurement of pH and pKa (Hydrogen, Quinhydrone, Glass electrodes) etc.

**MODULE 1 SECOND AND THIRD LAWS OF THERMODYNAMICS, FREE ENERGY, COLLIGATIVE PROPERTIES 12 Hrs**

Concept of entropy, entropy change in reversible process and irreversible process, free energy and entropy of mixing of ideal gas, ideal solutions, mixture of ideal solutions, Clausius inequality, Maxwell relations, Gibbs and Helmholtz energy, chemical potential.
Colligative properties: Relations between the four colligative properties boiling point, freezing point, solubility and osmosis

**MODULE 2 CHEMICAL EQUILIBRIUM, PHASE EQUILIBRIUM 12 Hrs**

Chemical equilibrium: Equilibrium constant of reaction of ideal gas and its temperature dependence: van't Hoff equation, Le Chatelier principle and application to the synthesis of NH3 (Haber's process) and SO3(Contact process).
Phase equilibrium: Phase transitions of pure substances: phase stability, boundary and phase rule, examples of phase diagrams; thermodynamics of phase transition: Clausius-Clapeyron equation, order of phase transition; phase diagrams of binary systems.

**MODULE 3 CHEMICAL KINETICS AND CATALYSIS 12 Hrs**

Chemical Kinetics: Extent of reaction, order of a chemical reaction, rate equations, differential and integrated rate laws: first, second and nth order reactions; unimolecular reactions, chain and photochemical reactions; Approximate methods: quasi steady state approximation; Arrhenius equation, energy of activation;
Homogeneous catalysis: acid-base catalysis; enzyme catalysis;
Heterogeneous catalysis: Adsorption and desorption of molecules, physisorption and chemisorption, Langmuir Isotherm.

**MODULE 4 ELECTROCHEMISTRY-I 12 Hrs**

Electrochemistry: conductivity of electrolytic solution, ionic mobility, migration of ions, strong and weak electrolytes, conductometric titrations, transport number and measurement, Grotthuss mechanism, ion activities and Debye-Hückel theory (qualitative descriptions only).
Electrochemical Cells: Daniell Reversible and irreversible cells, cell representations and half-cell reactions, Thermodynamics of electrochemical systems: Nernst equations, varieties of electrodes, standard electrode potential.

**MODULE 5 ELECTROCHEMISTRY-II** **12 Hrs**

Type of boundary between half cells and Liquid junction potentials, Concentration cells, Applications of EMF measurements-potentiometric titrations, determination of activity coefficient, composition of complex ions, solubility product, measurement of pH and pKa (Hydrogen, Quinhydrone, Glass electrodes), Polarization, Overvoltage.

**List of Experiments**

| S.no | Topic | Type |
|---|---|---|
| 1 | Molecular weight of a polymer (viscometry) | Experiment |
| 2 | Conductometric titrations (acid-base reaction) | Experiment |
| 3 | Phase diagram of a 2-component system | Experiment |
| 4 | pKa of amino acid (pH titration) | Experiment |
| 5 | Kinetics: Rate constant of acid catalyzed ester hydrolysis | Experiment |
| 6 | Partition co-efficient | Experiment |
| 7 | Conductometric titrations (precipitate reaction) | Experiment |

**Texbook(s):**

1. Ira N. Levine, Physical Chemistry, 6th, McGraw-Hill Education, 2007 ,9780071276368
2. Gilbert William Castellan, Physical Chemistry, 3rd, Addison-Wesley Publishing Company, 1983 ,9780201103861
3. Thomas Engel, Philip Reid, Physical Chemistry: Pearson New International Edition, 3rd, Pearson, 2013 ,9781292022246
4. Peter Atkins, Julio de Paula, Atkins' Physical Chemistry , 10th, Oxford University Press, 2014 ,9780199697403

**Reference(s):**

1. Robert J. Silbey, Robert A. Alberty, Moungi G. Bawendi, Physical Chemistry, 4, 2004 ,9780471215042
2. Donald A. McQuarrie, John D. Simon, Physical Chemistry: A Molecular Approach, 1, 1997 ,9780935702996

**Course Outcomes:**

1. Gain insights into ionic interactions in solutions and their role in solubility and conductance
2. Understanding rate of reaction and kinetics of reactions followed by catalysis
3. To provide insights into the concept of chemical equilibrium and factors influencing equilibrium
4. Measurement of pH and pKa (Hydrogen, Quinhydrone, Glass electrodes)
5. To introduce measurement of EMF-Nernst equation-effect of complexation on electrode potential.

**Course Articulation Matrix:**

| | POs | | | | | | | | | | | | | | | | PSOs | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **1** | **2** | **3** | **4** |
| **1** | 2 | 2 | 3 | | 2 | 3 | 2 | | 2 | 3 | 2 | | 3 | 2 | 2 | 2 | | 2 | | 3 |
| **2** | 2 | | 3 | 2 | 1 | 2 | | 2 | 3 | 3 | | 1 | 2 | 2 | 2 | 2 | | 2 | 3 | |
| **3** | 1 | 2 | 2 | 2 | 1 | 3 | 2 | 1 | 2 | 2 | 2 | 1 | 3 | | | 2 | | 3 | 2 | 1 |
| **4** | 3 | 2 | | | 2 | 3 | 2 | 3 | 2 | | 2 | | 3 | 2 | 2 | 3 | | 2 | | 2 |

| | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **5** | 3 | 1 | 1 | 3 | 2 | 2 | 1 | 3 | 1 | 1 | 3 | 2 | 2 | 1 | 1 | 2 | | 2 | 2 | 3 |

3 – High, 2 – Medium & 1 – Low Correlation

**APPROVED IN MEETINGS HELD ON:**

BOS : 26-06-2023 Academic Council Number: 27 Academic Council : 06-07-2023

**SDG No(s). & Statement(s) :**

4 & Quality Education : Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

**SDG Justification(s):**

The modules and topics mentioned in this course are designed to ensure all-inclusive and thorough education with equity to all persons and promote learning opportunities at all times.